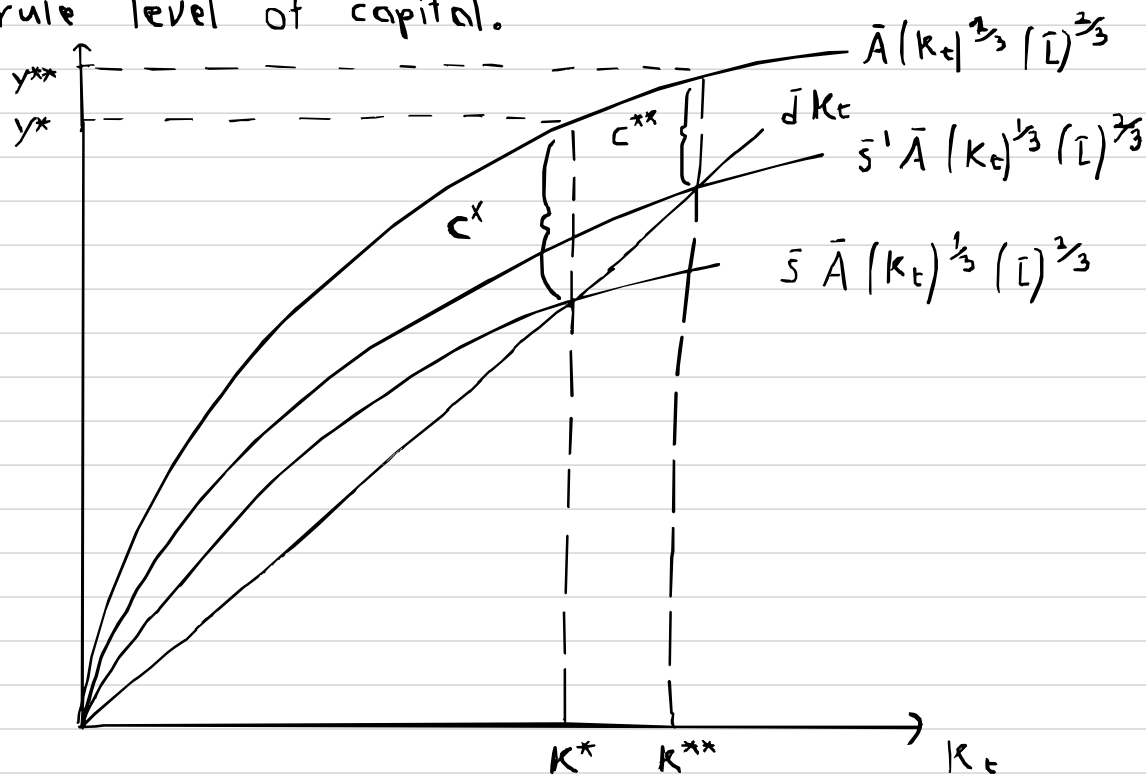


ECON 311 - Midterm Answer Key

Part I: Short-Answer Questions

Q1: Solow Model

The shock that best describes the dynamics in Figure 1 is an increase in savings at t_{shock} . In this case, k^* is above the golden-rule level of capital.



QII: Romer Model

$$i) \quad Y_t = A_t (1 - \bar{i}) \bar{L}$$

$$y_t = A_t (1 - \bar{i})$$

By the laws of growth:

$$g_{y_t} = g_{A_t} + g_{(1-\bar{i})} = g_{A_t}$$

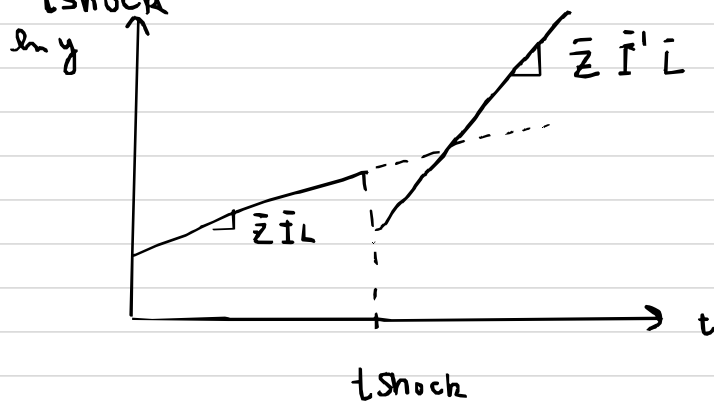
From productivity, we know:

$$\frac{A_{t+1} - A_t}{A_t} = g_{A_t} = \bar{z} \bar{i} \bar{L}$$

$$\therefore g_y = g_A = \bar{z} \bar{i} \bar{L}$$

ii)

$I_t \uparrow \Rightarrow g_y \uparrow$, but $y_t \downarrow$ because there are less workers available to produce at t . However, this means that future production per person increases because g_y is higher. This can be seen in the following graph where we assume that $I \uparrow$ to $\bar{i}'$ at t_{shock}



Q III : Economic Growth

(a)

$$\bar{g}_{ARG} = \left(\frac{20,222}{2,890} \right)^{\frac{1}{64}} - 1 = 0.0309$$

$$\bar{g}_J = \left(\frac{35,358}{2,616} \right)^{\frac{1}{64}} - 1 = 0.0415$$

(b)

$$T = \frac{\ln \left(\frac{35,358}{20,222} \right)}{0.0309} = 18.0826$$

(c)

$$y_{2024}^{ARG} = y_{2014}^{ARG} (1 + 0.031)^{10} = 27,441.68$$

$$y_{2014}^{Turkey} = \frac{y_{2024}^{Turkey}}{(1.015)^{10}} = \frac{27,441.68}{(1.015)^{10}} = 23,636.24$$

Part II: Long-Answer Questions

Q1: Model of Production

(a)

$$\begin{aligned} F(\lambda K_t, \lambda L_t, \lambda N_t) &= \bar{A} (\lambda K_t)^{\frac{1}{4}} (\lambda L_t)^{\frac{1}{2}} (\lambda N_t)^{\frac{1}{4}} \\ &= \lambda^{\left(\frac{1}{4} + \frac{1}{2} + \frac{1}{4}\right)} \bar{A} K_t^{\frac{1}{4}} L_t^{\frac{1}{2}} N_t^{\frac{1}{4}} \\ &= \lambda F(K_t, L_t, N_t) \end{aligned}$$

$\therefore$ It exhibits constant returns to scale

(b)

$$g_y = g_{\bar{A}} + \frac{1}{4} g_K + \frac{1}{2} g_L + \frac{1}{4} g_N$$

$$\Rightarrow g_y = \frac{3}{4} g > 0$$

$$\frac{y_t}{L_t} = \bar{A} \left(\frac{K_t}{L_t} \right)^{\frac{1}{4}} \left(\frac{N_t}{L_t} \right)^{\frac{1}{4}}$$

$$g_y = -\frac{1}{4} g < 0$$

Production depends positively on K & L , so when both increase, so does production. However, y exhibits diminishing returns to both K, L , so when we divide it by L , the denominator grows faster than the numerator.

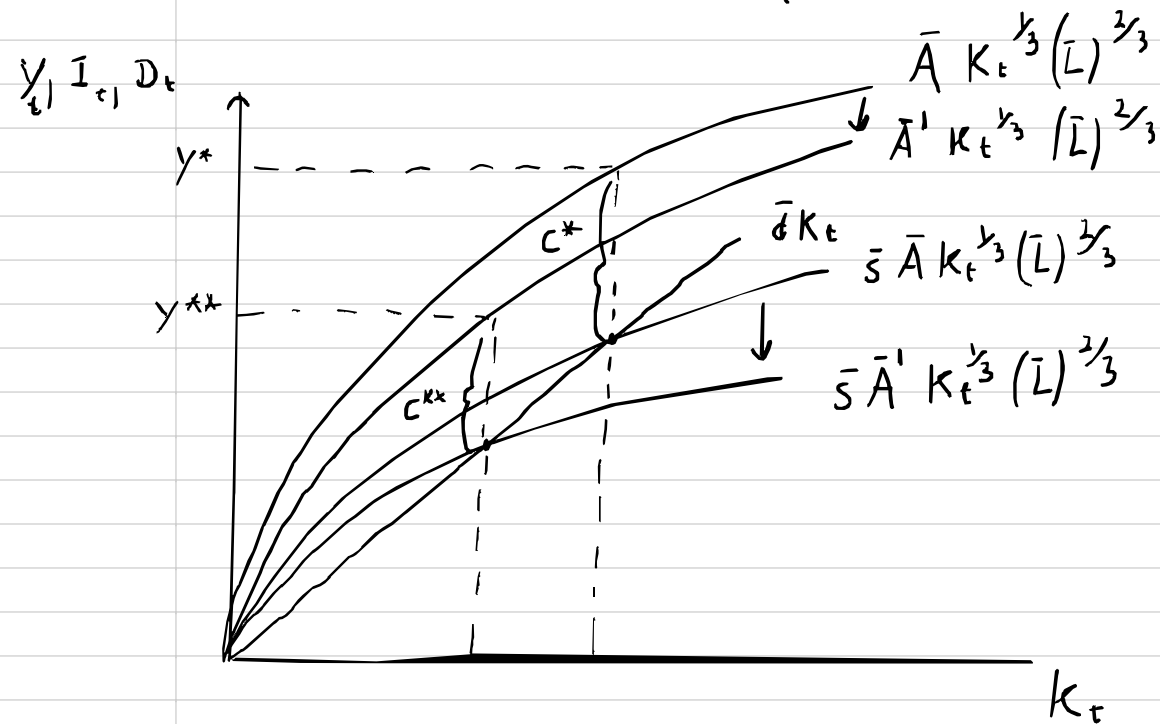
(c)

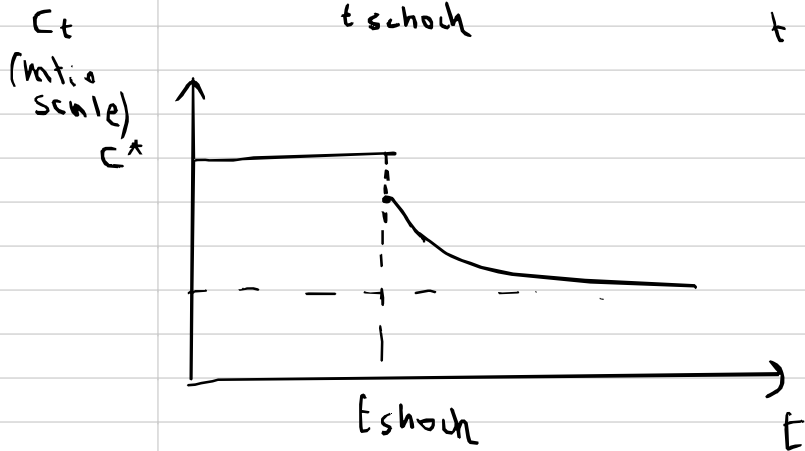
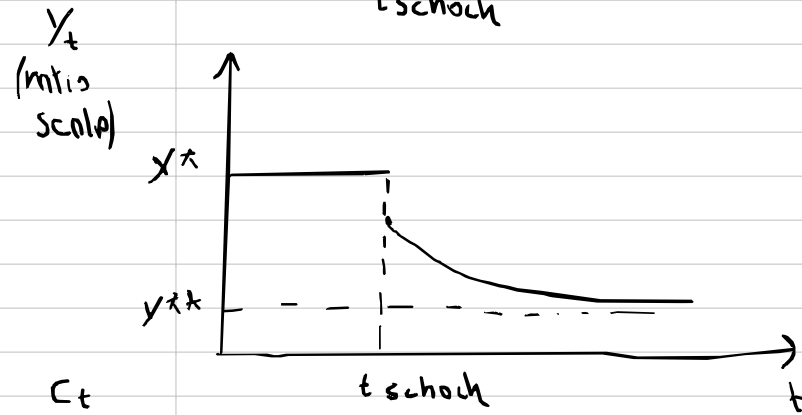
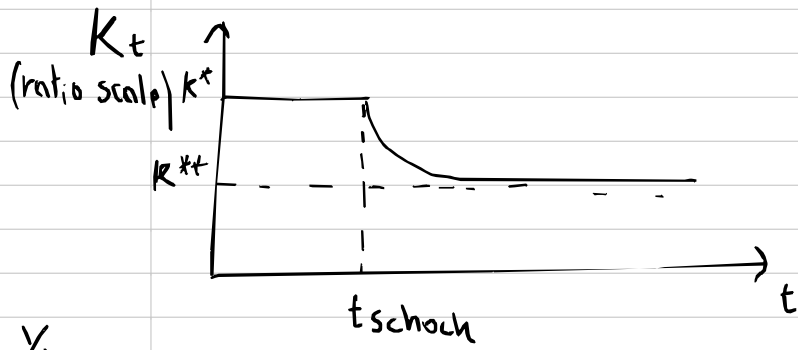
$$g_y = \frac{7}{4} g > 0$$

$$g_y = \frac{3}{4} g > 0$$

When A also grows, both y and $\frac{y}{L}$ have positive growth. The reason for this is that the growth of y now exceeds the growth of L .

QII: Solow Growth Model





QIII: Measuring the Macroeconomy

$$NGDP_{2017} = 1,000 + 500 \cdot 15 = 8,500$$

$$NGDP_{2018} = 1,100 \cdot (1.5) + 700 \cdot (16.5) = 13,200$$

$$RGDP_{2017}^{\$2017} = NGDP_{2017} = 8,500$$

$$RGDP_{2018}^{\$2017} = (1,100) + 700 \cdot (15) = 11,600$$

$$\Delta\% RGDP = 0.3647$$

Note that:

$$8,500 \times 1.3647 = 11,600$$

$$GDP_{def} 2017 = 1$$

$$GDP_{def} 2018 = \frac{13,200}{11,600} = 1.1379$$

So, inflation = 13.79 %
rate

Changing the benchmark does not affect the result.